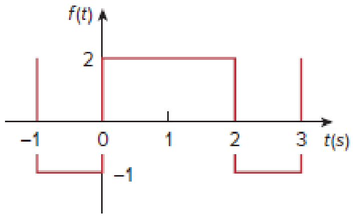
Chapter 17, Problem 7P

Problem

Determine the Fourier series of the periodic function in Fig. 17.50.

Figure 17.50:



Step-by-step solution

Step 1 of 1

From the given wave form we have

$$f(t) = 2 \text{ for } 0 < t < 2$$
$$f(t) = -1 \text{ for } 2 < t < 3$$

We have

$$T = 3$$

$$\omega_0 = \frac{2\pi}{T}$$

$$\therefore \omega_0 = \frac{2\pi}{3}$$

We know,

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt$$

$$a_0 = \frac{1}{3}\left[\int_0^2 2dt + \int_2^3 -1dt\right]$$

$$a_0 = \frac{1}{3}\left[[2t]_0^2 - [t]_2^3\right]$$

$$a_0 = \frac{1}{3}\left[[2(2-0)] - [3-2]\right]$$

$$a_0 = \frac{1}{3}[4-1]$$

$$a_0 = \frac{1}{3}[3]$$

$$\therefore a_0 = 1$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt$$

$$a_n = \frac{2}{3}\left[\int_0^2 2\cos n\omega_0 t\,dt + \int_2^3 -1\cos n\omega_0 t\,dt\right]$$

$$a_n = \frac{2}{3}\left[\int_0^2 2\cos n\omega_0 t\,dt - \int_2^3 \cos n\omega_0 t\,dt\right]$$

$$a_n = \frac{2}{3}\left\{\left[\frac{2\sin n\omega_0 t}{n\omega_0}\right]_0^2 - \left[\frac{\sin n\omega_0 t}{n\omega_0}\right]_2^3\right\}$$

$$a_n = \frac{2}{3}\left\{\left[\frac{2\sin 2n\omega_0}{n\omega_0}\right] - \left[\frac{\sin 3n\omega_0}{n\omega_0} - \frac{\sin 2n\omega_0}{n\omega_0}\right]\right\}$$

$$a_n = \frac{2}{3}\left\{\frac{2\sin 2n\omega_0}{n\omega_0} - \frac{\sin 3n\omega_0}{n\omega_0} + \frac{\sin 2n\omega_0}{n\omega_0}\right\}$$

$$a_n = \frac{2}{3}\left\{\frac{3\sin 2n\omega_0}{n\omega_0} - \frac{\sin 3n\omega_0}{n\omega_0}\right\}$$

$$a_n = \frac{2}{3n\omega_0}\{3\sin 2n\omega_0 - \sin 3n\omega_0\}$$

$$a_n = \frac{2}{3n\left(\frac{2\pi}{3}\right)}\left\{3\sin 2n\left(\frac{2\pi}{3}\right) - \sin 3n\left(\frac{2\pi}{3}\right)\right\}$$

$$a_n = \frac{1}{n\pi}\left\{3\sin\left(\frac{4\pi n}{3}\right) - \sin 2\pi n\right\}$$

$$\therefore a_n = \frac{1}{n\pi}\left\{3\sin\frac{4\pi n}{3}\right\} \qquad (\because \sin 2\pi n = 0)$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt$$

$$b_n = \frac{2}{3}\left[\int_0^2 2\sin n\omega_0 t\,dt + \int_2^3 -1\sin n\omega_0 t\,dt\right]$$

$$b_n = \frac{2}{3}\left[\int_0^2 2\sin \omega_0 t\,dt - \int_2^3 \sin n\omega_0 t\,dt\right]$$

$$b_n = \frac{2}{3}\left\{\left[-\frac{2\cos n\omega_0 t}{n\omega_0}\right]_0^2 - \left[-\frac{\cos n\omega_0 t}{n\omega_0}\right]_2^3\right\}$$

$$b_n = \frac{2}{3}\left\{\left[-\frac{2\cos 2n\omega_0}{n\omega_0} + \frac{2}{n\omega_0}\right] + \left[\frac{\cos 3n\omega_0}{n\omega_0} - \frac{\cos 2n\omega_0}{n\omega_0}\right]\right\}$$

$$b_n = \frac{2}{3}\left\{-\frac{3\cos 2n\omega_0}{n\omega_0} + \frac{2}{n\omega_0} + \frac{\cos 3n\omega_0}{n\omega_0}\right\}$$

$$b_n = \frac{2}{3n\omega_0}\{-3\cos 2n\omega_0 + 2 + \cos 3n\omega_0\}$$

$$b_n = \frac{2}{3n\left(\frac{2\pi}{3}\right)}\left\{-3\cos 2n\left(\frac{2\pi}{3}\right) + 2 + \cos 3n\left(\frac{2\pi}{3}\right)\right\}$$

$$b_n = \frac{1}{n\pi}\left\{-3\cos\left(\frac{4\pi n}{3}\right) + 2 + \cos 2\pi n\right\}$$

$$b_n = \frac{1}{n\pi}\left\{-3\cos\left(\frac{4\pi n}{3}\right) + 2 + 1\right\} \qquad (\because \cos 2\pi n = 1)$$

$$b_n = \frac{1}{n\pi}\left\{3 - 3\cos\left(\frac{4\pi n}{3}\right)\right\}$$

$$\therefore b_n = \frac{3}{n\pi}\left\{1 - \cos\frac{4\pi n}{3}\right\}$$

Hence the Fourier series is

$$f(t) = 1 + \sum_{n=0}^{\infty}\left[\frac{3}{n\pi}\sin\frac{4\pi n}{3}\cos\frac{2n\pi t}{3} + \frac{3}{n\pi}\left(1 - \cos\frac{4\pi n}{3}\right)\sin\frac{2n\pi t}{3}\right]$$